

Open X-Embodiment · arXiv 2310.08864 · Nov 2 · Data

Visionary

11+3 · seat 1 = Kanta · HIGH

OXE = municipal water · BEHAVIOR'26 = bottled blend
Steal mix science — don't drink the river

Paper club · Data cluster · due 9:30 AM PT

Spine: pool $\neq$ train mix · capacity $\times$ mix · steal mix science, don't drink the river

Open X-Embodiment Collaboration · 21 institutions · robotics-transformer-x.github.io

Visionary talk arc · 11+3



Spine: pool $\neq$ train mix · capacity×mix · steal mix science, don't drink the river

Visionary map — what to take into 2026

ONE-LINER

OXE = municipal water for generalist robot policies;
BEHAVIOR'26 = bottled specialty blend — steal mix science, don't drink the river.

Steal

Width · Schema · Lang column
Transfer ablations · Cap×mix

Fix

Weight table · Kill switch
Wrist gate · $n^{0.43}$ · chunks

Ship

$\pi_{0.5}$ / N1.7 substrates
4-stage mix · holdout science

No Discord/Drive · paper-grounded · local quiet pack only

CLEAR: Pool ≠ RT-X train mix

Marketing blur is the #1 citation failure mode

OXE POOL

Datasets	60
Embodiments	22
Institutions / labs	21 / 34
Trajectories	1M+
Skills / tasks	527 / 160,266
Scenes	311
Schema	RLDS TFRecords

RT-X TRAIN MIX

Manipulators	9 (≠ full 22)
Sources (ex.)	RT-1, QT-Opt, Bridge, Play, Jaco, Cable, ...
RT-1-X	robotics mix only
RT-2-X	~1:1 web : robotics
Eval	3600 trials / 6 robots
Downstream	Octo ~800k · OpenVLA ~970k

Moral: every claim needs a frozen mix card — pool size ≠ train size ≠ Octo/OpenVLA subset

Cite “22 robots / 1M+” for the commons; cite “9 manipulators” for RT-X experiment results

Steal from OXE — five first-class lessons

Visionary · seat 1 = Kanta · BEHAVIOR'26 data design

1

Width

Embodiment / scene / object / skill
coverage as first-class objective

2

Schema before scale

RLDS → lock obs keys early
(wrist, proprio, lang, subgoals)

3

Language as mix column

Not a garnish — push post-train
lang coverage >90% + paraphrases

4

Transfer ablations

Bridge-removal is gold standard
Publish negative transfer too

5

Capacity × mix

Table I: small policies underfit
rich mixes — co-design both

OXE proved agglomeration works · Octo/OpenVLA/ π_0 proved you still have to cook the soup

For BEHAVIOR: steal coverage + schema + transfer ablations — not the RT-X training bill

Anti-patterns → fixes for BEHAVIOR

OXE-era raw practice → later soup science → your action

OXE-era fail	Fix to copy	BEHAVIOR action
Uniform everything	Curated weight table	Explicit mix card
Loyalty to bad shards	Kill switch → weight 0	Drop when loss stalls
Wrist poverty	Hard gate contact tasks	No wrist → down-weight
Lang poverty	Dual-condition + subgoals	Lang $\geq 90\%$ post-train
7×256 bins only	Chunked continuous / FAST	Not naïve discrete EE
Huge shards drown gold	Sublinear $n^{0.43}$	Protect $\sim 20k$ gold demos
Pick-place monopoly	Skill-family caps	Cap family $\leq \sim 20\%$

Concrete 4-stage BEHAVIOR mix recipe

Prior → Mid-train sim → Post-train gold → Holdout mix science

1

PRIOR

OXE-like / vendor $\pi_{0.5}$ /GR00T
Sublinear weights
Exclude bad shards
Goal: semantic + visuomotor breadth

2

MID-TRAIN SIM

Cover all ~100 tasks
Upsample long-horizon +
failure recovery
Dense lang / subtask tokens

3

POST-TRAIN GOLD

Highest sample weight
Per-task floors
Wrist-complete demos
BEHAVIOR tokens ≥ 30 -50%
of late training

4

HOLDOUT MIX SCIENCE

Ablate: no-prior / no-wrist
no-paraphrase / uniform vs
 $n^{0.43}$ / kill switch
Publish transfer + negatives

Winning pattern: large diverse prior → small high-quality specialized finetune

Red lines + municipal water metaphor

OXE = MUNICIPAL WATER

Shared multi-lab supply
Enabled RT-X / Octo / OpenVLA / π_0
Agglomeration + RLDS schema
Positive transfer existence proof

BEHAVIOR'26 = BOTTLED BLEND

Specialty challenge mix
Stage-rich · wrist-rich · schema-clean
~20k gold demos as post-train depth
Steal science — don't drink the river

DO NOT

- × Dump raw OXE into BEHAVIOR FT at uniform weight
- × Assume discrete 7×256 bins are enough for 2026 long-horizon household
- × Claim “OXE alone = BEHAVIOR'26 data plan”
- × Cite unpublished OXE hours
- × Skip mix-card discipline / transfer ablations

BEHAVIOR mix checklist (steal-this)

☐ Weight table checked in (dataset → sample prob)

☐ Language coverage $\geq 90\%$ of post-train

☐ Wrist coverage tracked; contact tasks gated

☐ Skill-family caps (anti pick-place monopoly $\leq \sim 20\%$)

☐ Sublinear size temperature (protect $\sim 20k$ gold)

☐ Kill switch for underfitting shards

☐ Holdout task-family transfer ablation

☐ Sim/real columns separated

☐ Action = chunked continuous (or FAST), not naïve 7×256 only

☐ CLEAR / TENTATIVE labels on OXE-derived claims

Headline numbers (tattoo)

Fig. 4

~+50%

relative mean SR lift
small-data domains
RT-1-X vs Original / RT-1

Table I

73% vs 92%

RT-1-X underfits
Google Robot 6 skills
capacity × mix

Table II

27.3 → 75.8

emergent skills ~3×
Bridge→Google Robot
X-mix transfer

Ablation

42.8%

no-Bridge mix
emergent collapses
Bridge is load-bearing

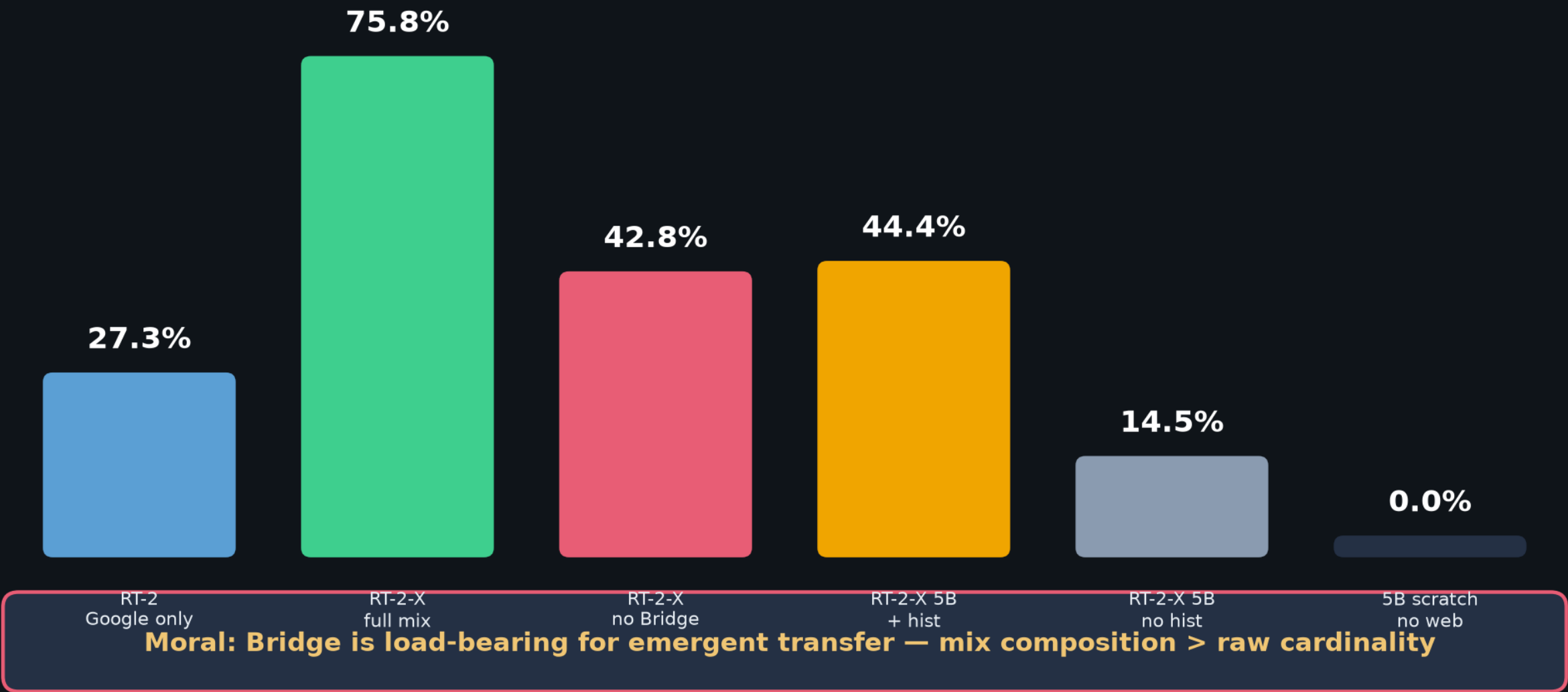
Table I — large native domains (exact)

Setting	Original	RT-1	RT-1-X	RT-2-X 55B
Bridge IRIS WidowX	13%	40%	27%	50%
Bridge RAIL WidowX	13%	30%	27%	30%
Google 6 skills	—	92%	73%	91%

RT-1-X hurts large in-domain SR — positive transfer is conditional on capacity

Table II — Bridge ablation (emergent skills)

Bridge→Google Robot skills absent from Google-only data



Capacity × mix co-design (Table I moral)

More data is not free — small models underfit rich X-mixes on large native domains

RT-1-X (35M) on X-mix

Google 6-skill: 73% (vs RT-1 92%)

Bridge IRIS: 27% (vs RT-1 40%)

Underfits large native sets

RT-2-X (55B) on X-mix

Google 6-skill: 91% (recovers)

Bridge IRIS: 50% (beats)

Capacity absorbs diversity

BEHAVIOR implication

- If shipping $\pi_{0.5}$ / GROOT-scale → you can afford OXE-like width (with weight table).
- If shipping a tiny policy → curate narrower; gate external shards behind validation q-score.
- Empiricist underfit probe: same heavy mix, small vs base open ckpt, matched steps.

Genealogy — OXE → soups → BEHAVIOR



What OXE replaced

Single-lab teleop as default pretrain
“One robot, one policy” as only recipe

What OXE did NOT replace

Post-train gold demos · wrist/proprio
Continuous chunks · challenge sim alignment

Children drink the water: Octo compares to RT-1-X · OpenVLA beats RT-2-X on open suite

π_0 open share $\approx$ 9.1% OXE Magic Soup + Bridge + DROID · BEHAVIOR needs OXE-scale thinking, not raw dump

Fig. 4 — Small-data domains: RT-1-X ~+50% mean lift

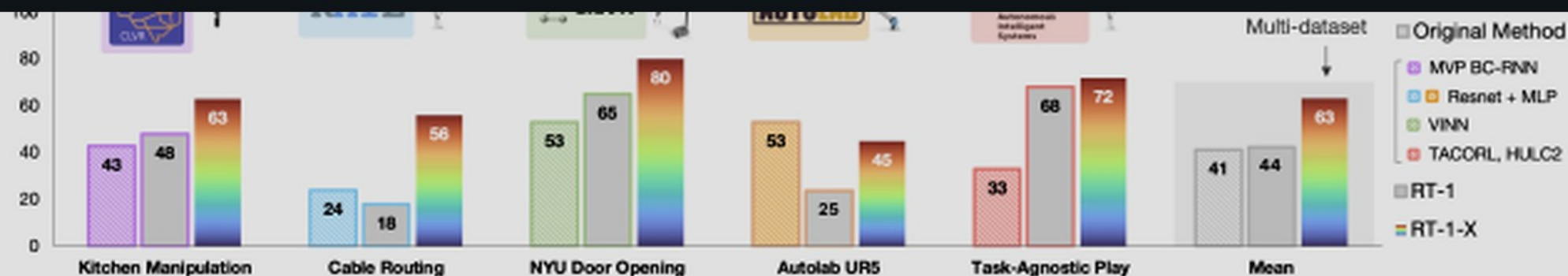


Fig. 4: RT-1-X mean success rate is 50% higher than that of either the Original Method or RT-1. RT-1 and RT-1-X have the same network architecture. Therefore the performance increase can be attributed to co-training on the robotics data mixture. The lab logos indicate the physical location of real robot evaluation, and the robot pictures indicate the embodiment used for the evaluation.

Evaluation Setting	Bridge	Bridge	RT-1 paper 6 skills
Evaluation Location	IRIS (Stanford)	RAIL Lab (UCB)	Google Robotic Lab
Robot Embodiment	WidowX	WidowX	Google Robot
Original Method	LCBC [95]	LCBC [95]	-
Original Method	13%	13%	-
RT-1	40%	30%	92%
RT-1-X	27%	27%	73%
RT-2-X (55B)	50%	30%	91%

TABLE I: Parameter count scaling experiment to assess the impact of capacity on absorbing large-scale diverse embodiment data. For these large-scale datasets (Bridge and RT-1 paper data), RT-1-X underfits and performs worse than the Original Method and RT-1. RT-2-X model with significantly many more parameters can obtain strong performance in these two evaluation scenarios.

tasks. We split our evaluation into two types: evaluation on domains that have small-scale datasets (Fig. 4), where we would expect transfer from larger datasets to significantly improve performance, and evaluation on domains that have large-scale datasets (Table I), where we expect further improvement to be more challenging. Note that we use the same robotics data *training* mixture (defined in Sec. IV-C) for all the evaluations presented in this section. For small-scale dataset experiments, we use Kitchen Manipulation [128], Cable Routing [129], NYU Door Opening [130], AUTOLab UR5 [132], and Robot Play [134]. We use the same evaluation and robot embodiment as in the respective publications. For large-scale dataset experiments, we consider Bridge [95]

Visionary · close

Steal mix science · cook a curated soup **Don't drink raw OXE as BEHAVIOR plan**

Width · schema · lang column · transfer ablations · capacity×mix

4-stage: Prior → Mid-sim → Post-gold → Holdout mix science

No Discord/Drive

Q&A · keep the spine clear

Open X-Embodiment

Robotic Learning Datasets and RT-X Models

arXiv 2310.08864 · Nov 2 Data cluster

OXE = municipal water · BEHAVIOR'26 = bottled specialty blend

Steal mix science — don't drink the river

Pool: 60 ds / 22 emb / 1M+ traj Train mix: 9 manipulators Eval: 3600 trials / 6 robots

Headlines: Fig.4 ~+50% · Table I RT-1-X 73% vs RT-1 92% · emergent 27.3%→75.8% · no-Bridge 42.8%

Seat 1 Visionary = Kanta Saito · Quiet build